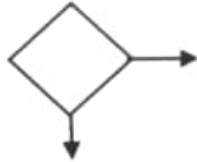
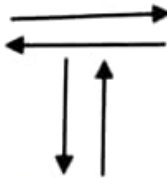


1.3.1 Flowcharts

Flowcharts are used to present how the algorithm is built step by step in a dramatic manner. The symbols given in table are used to indicate different functions. (See Table 1.1)

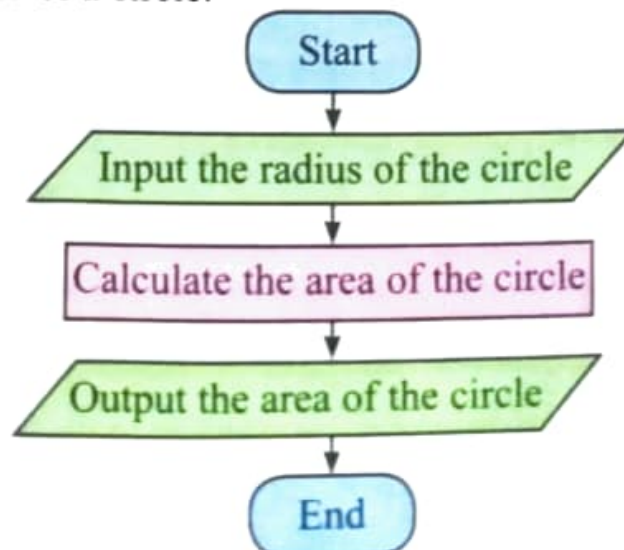
Table 1.1

Symbol	Function
	Start or end
	Input or output
	Process
	Decision
	Flow direction
	Connector

Sequence

In sequence the steps from the beginning to the end are executed in order.

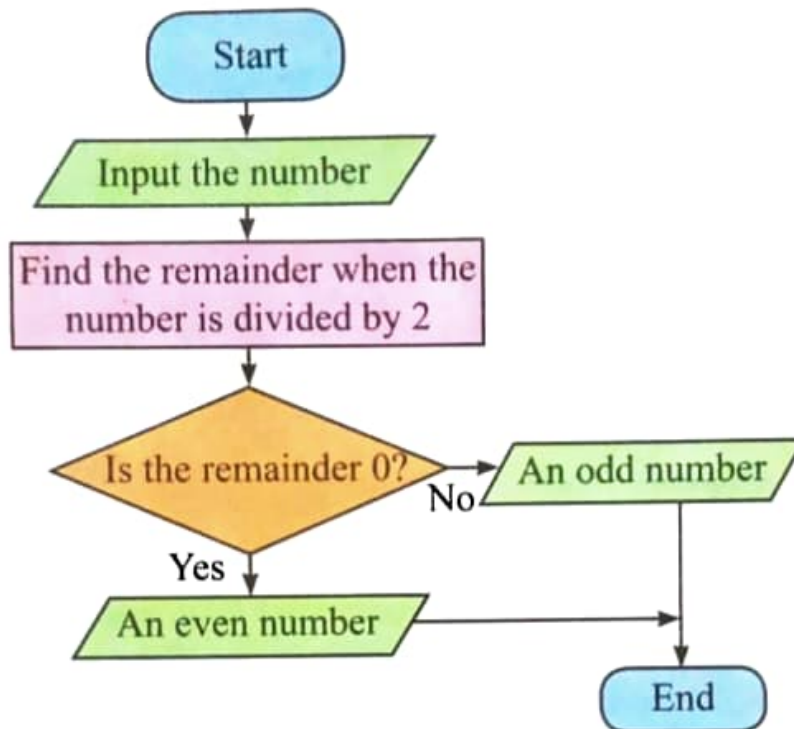
E.g. 1 - Finding the area of a circle.



Selection

The selection indicates the flow of direction depending on a condition being satisfied or not.

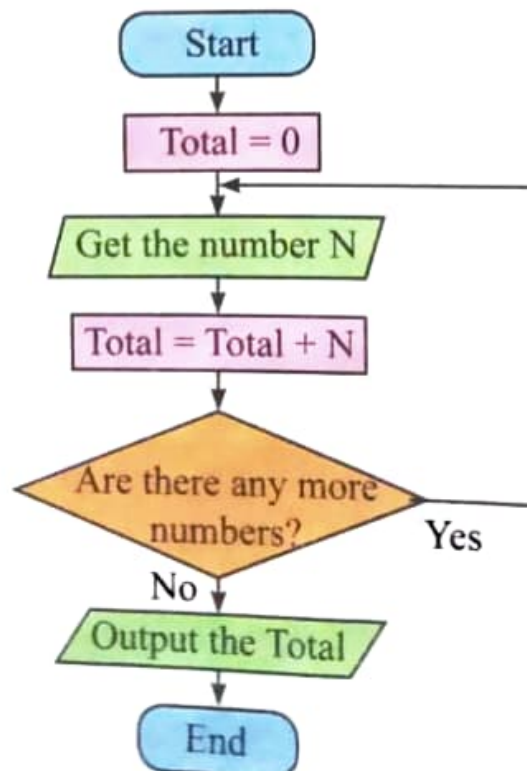
E.g. - Finding whether a number is odd or even



Repetition

A step or several steps are repeated till a condition is satisfied or are continued till it is satisfied.

E.g. - Finding the total of some numbers



Activity



Draw flow charts to solve the problems given below.

1. Find the perimeter and area of a rectangle.
2. It is decided to add Rs.5000 to the basic salary of the employees of a company. Calculate the new salary.
3. When posting a letter, postal fare is charged according to its weight. The standard fare should be paid for letters which are equal to or less than the standard weight. An additional fare should be charged if the weight is more than the standard weight.
4. Indicate the first 12 multiples of 7.
5. Draw flow charts for the examples in selection 1:2:1.

1.3.2 Pseudo codes

When an algorithm is presented in simple English terms it is called a pseudo code. Pseudo codes are independent of a computer language. Pseudo codes can be converted to any programming language instructions. Hence, pseudo codes make computer programming easier.

Let us see simple English terms used in an pseudo code.

BEGIN - To indicate a beginning

END - To indicate an end

INPUT , READ , GET - To indicate an input

OUTPUT, DISPLAY , SHOW - To show an output

PROCESS, CALCULATE - To indicate a process

IF ... THEN ...ELSE ... ENDIF - Used to indicate a selection

FOR - DO

WHILE - ENDWHILE

REPEAT - UNTIL

} Used to indicate a repetition

Writing pseudo codes

E.g. 1 - Finding the area of a circle

BEGIN

INPUT Radius

CALCULATE Area = $22/7 \times \text{Radius} \times \text{Radius}$

DISPLAY Area

END.

E.g. 2 - Finding whether a number is odd or even

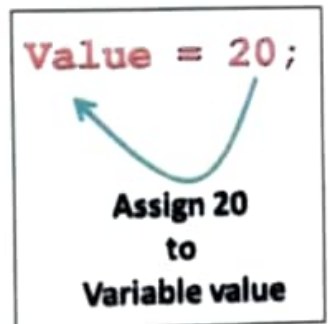
```
BEGIN
  READ number as N
  CALCULATE Remainder after number is divided by 2
  IF Remainder = 0 THEN
    DISPLAY "Even number"
  ELSE
    DISPLAY "Odd number"
  ENDIF
END.
```

E.g. 3 - Finding the total of some numbers

```
BEGIN
  Total = 0
  REPEAT
    READ Number as N
    CALCULATE Total = Total + N
  UNTIL numbers are over
  DISPLAY Total
END.
```

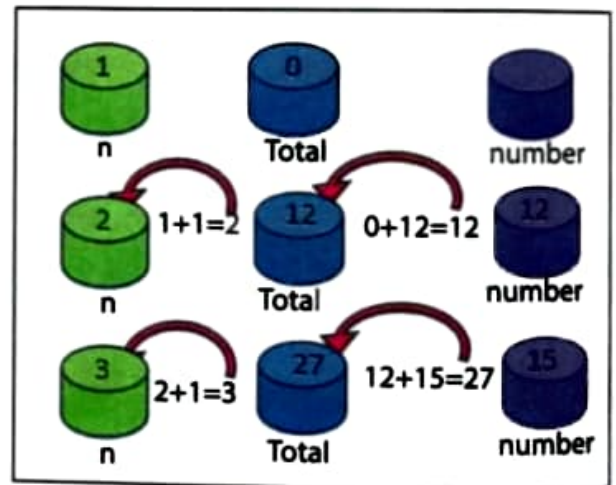
E.g. 4 - Finding the total and the average of 10 numbers

```
BEGIN
  Total = 0
  Average = 0
  n = 1
  WHILE n <= 10
    READ Number
    CALCULATE Total = Total + Number
    n = n + 1
  ENDWHILE
  CALCULATE Average = Total/(n - 1)
  DISPLAY Total, Average
END.
```



Following are some of the facts about the above pseudo code.

- Total, Average and Number are variables.
 - When values are assigned to number variable, the value of Total, Average and n variables change.
- n indicates the number of repetitions. (Number of times the loop is executed)
- The statements $\text{Total} = 0$ and $\text{Average} = 0$, makes starting values of these variables assigned as 0.
 - Hence, the initial value of Total and Average are 0s.
- The statement $n = 1$, makes the starting value of n is assigned to 1.
- $n \leq 10$ is the condition that should be satisfied.
- WHILE $n \leq 10$ indicates that the loop n should be repeated until value of n is 10.
 - Repetition occurs when the value of n is 10 or less than 10. This means, till the condition $n \leq 10$ is true, repetition occurs. When the value of n becomes 11, the repetition stops. Then the condition becomes false.
- READ denotes getting a value for Number variable.
- $\text{Total} = \text{Total} + \text{Number}$ denotes the present value of Total is added to number and the resultant value is assigned on the new value of Total.
- $n = n + 1$ calculates the number of repetitions. 1 is added to the present value of n and the result is assigned to n.
- ENDWHILE indicates the limit to end repetition. Hence, only READ number, $\text{Total} = \text{Total} + \text{Number}$ and $n = n + 1$ are repeated till the condition $n \leq 10$ is fulfilled.
- When the repetition stops, the value of n is 11 and the condition is false.
- By $\text{Average} = \text{Total} / (n - 1)$, final value of Total is divided by $(n - 1)$ and that value is assigned to Average variable.
- DISPLAY (Total, Average) produces the output of the total of 10 numbers and its average.



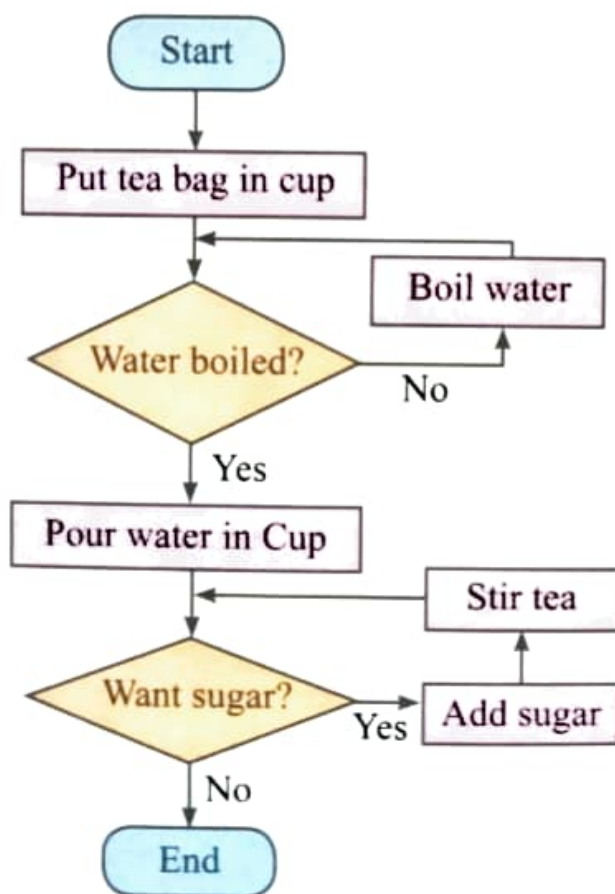
Observation



- When a value is assigned to a variable, the previous value is lost.
- When the statement $\text{Total} = \text{Total} + \text{Number}$ is executed, the value assigned to the Number variable is added with the value assigned to Total variable and the result obtained is assigned to the Total variable.
- $\text{Total} = \text{Total} + \text{Number}$ is not a mathematical formula.

1.3.3 Converting flow charts to pseudo codes

An algorithm can be presented in a flow chart as well as in a pseudo code. Hence, let us examine how a flow chart can be converted to a pseudo code.



BEGIN

Put tea bag in cup

WHILE (not water boiled)

Boil water

ENDWHILE

Pour water in cup

WHILE (sugar needed)

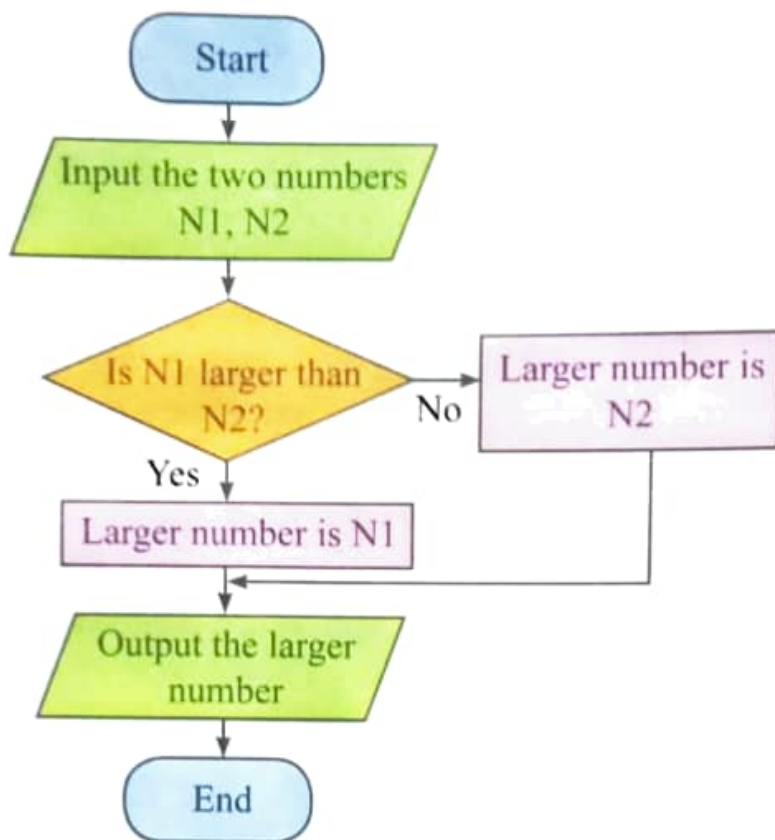
Add sugar

Stir tea

ENDWHILE

END

E.g. 1 - Finding the larger number from two different numbers



BEGIN

READ N1, N2

IF N1 > N2 THEN

Large = N1

ELSE

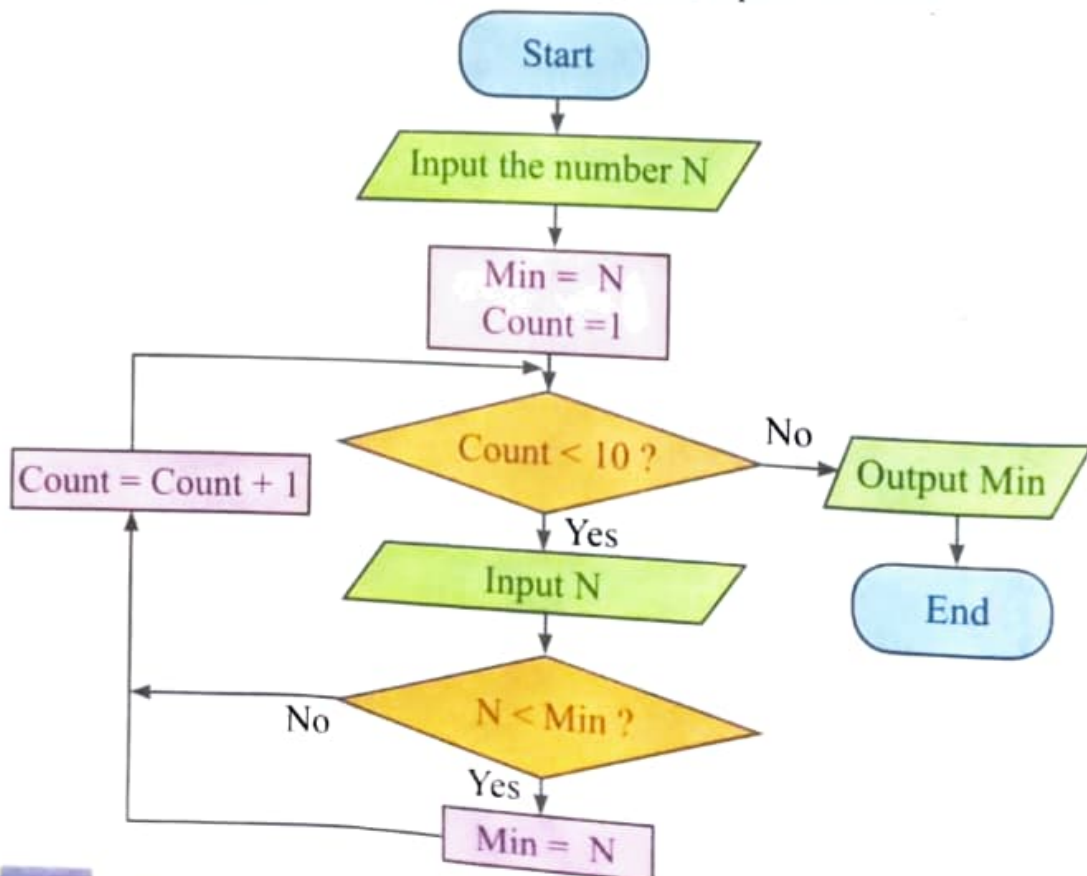
Large = N2

ENDIF

DISPLAY Large

End.

E.g. 2 - Finding the smallest number from 10 input numbers



BEGIN

INPUT Number as N

Min = N

Count = 1

WHILE Count < 10

OUTPUT Number as N

IF N < Min Then

Min = N

ENDIF

Count = Count + 1

ENDWHILE

PRINT Min

END.